

Example-1.

Using the Aggregation Framework, perform the following operations:

1. Group students based on their city.
2. Calculate the average marks of students in each city.
3. Sort the result in descending order of average marks.
4. Display only the city name and the average marks.

Example-2.

Using the Aggregation Framework, perform the following operations:

1. Select students whose marks are greater than or equal to 80.
2. Group the selected students by city.
3. Calculate the total marks obtained by students in each city.
4. Sort the result in descending order of total marks.
5. Display only the city name and total marks.

Example-3.

Create Collection orders and solve the question using the concept of aggregation.

```
db.orders.insertMany([
  {_id:1,    customer:"Nidhi",    product:"Laptop",    category:"Electronics",
  amount:55000},
  {_id:2,    customer:"Rahul",    product:"Mobile",    category:"Electronics",
  amount:25000},
  {_id:3, customer:"Priya", product:"Shoes", category:"Fashion", amount:3000},
  {_id:4, customer:"Amit", product:"T-Shirt", category:"Fashion", amount:1500},
  {_id:5,    customer:"Neha",    product:"Refrigerator",    category:"Electronics",
  amount:40000},
  {_id:6, customer:"Karan", product:"Watch", category:"Fashion", amount:5000},
  {_id:7,    customer:"Riya",    product:"Microwave",    category:"Electronics",
  amount:12000}
])
```

Question 1

Calculate the average order amount for each category. Sort the result in descending order of average amount and display only category and average amount.

Question 2

Calculate the total sales amount for each category. Sort the result in descending order of total sales and display only category and total sales.

Question 3

Find the highest order amount for each category. Sort the result in descending order of highest order amount and display only the category and highest order amount.

Question 4

Count the number of orders in each category. Display only the category and total number of orders.

Question 5

Find the minimum order amount for each category. Display only the category and minimum order amount.